Lab 03
LOGIC GATES & ADDERS

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3.1 Objectives

After performing this lab, student should be able to:
- o Verify the behavior of XOR logic gate
- o Utilize Datasheet to get familiarized with digital integrated circuits and the internal IC gate level structure.
- o Construct logic circuit of half Half-Adder.

3.2 Equipment/Apparatus

1. Adjustable Direct Current (DC) Power supply.
2. Digital Multi Meter (DMM).
3. XOR gate IC 7486.
4. Light Emitting Diode (LED)
5. Switches.
6. Resistors (330Ω)
7. AND gate IC 7408.

3.3 Reading Datasheet

Task a
In this task you are required to read the datasheet to get familiarized with internal gate level structure of XOR gate IC 7486.

3.4 XOR logic gate

An XOR gate implements an exclusive OR; that is, a true output results if one, and only one, of the inputs to the gate is true. If both inputs are false (0/LOW) or both are true, a false output result. XOR represents the inequality function, i.e., the output is true if the inputs are not alike.
The logic symbol or Boolean expression for an XOR gate is add sign in circle $'\oplus'$. It describes the XOR operation on two inputs. In the expression below, A and B are the inputs and Y is the output. Table 3.1 provides the truth table of XOR gate.

$$Y = A \oplus B$$

Three input XOR operation can be performed with the simple expression shown below; where A, B and C are inputs, and Y is the output.

$$Y = A \oplus B \oplus C$$

Table 3.1 - Truth table for the 2-input XOR gate

Input (A)	Input (B)	Output (Y)
0	0	0
0	1	1
1	0	1
1	1	0

Task b

ii. Procedures and observations

1. Fix the IC on the half separator line of breadboard, so there are no short connections.
2. Connect a wire to the main voltage source (Vcc = 5 V) whose another end is connected to last pin of the IC (14th pin from the notch).
3. Connect the ground pin of the IC (7^{th} pin from the notch) to the ground terminal provided by DC power supply.
4. Give the input at any one gate of the IC i.e., 1^{st}, 2^{nd}, 3^{rd} or 4^{th} gate by using connecting wires (in accordance to the datasheet of IC).
5. Connect output pins to an LED through a resistor.
6. Connect Voltmeters at inputs and output pins of the IC.
7. Switch on the power supply.
8. If LED glows then output is true, if it does not glow, the output is false, which is numerically denoted as logic 1 (high) and logic 0 (low) respectively. Test the truth table provided in Table 3.1.
9. Complete Table 3.2 with two inputs.
10. From the given IC, use two 2-input XOR gates to complete Table 3.3 with three inputs.

Table 3.2 - Data Table for 2-Input XOR Gate

Input				Output	
A (Logic)	B (Logic)	A (Volts)	B (Volts)	Y (Logic)	Y (Volts)
0	0	0	0	0	0
0	1	0	5	1	4.8
1	0	5	0	1	4.8
1	1	5	5	0	0

Table 3.3 - Data Table for 3-Input XOR Gate

Input						Output	
A (Logic)	B (Logic)	C (Logic)	A (Volts)	B (Volts)	C (Volts)	Y (Logic)	Y (Volts)
0	0	0	0	0	0	0	0
0	0	1	0	0	5	1	4.8
0	1	0	0	5	0	1	4.8
0	1	1	0	5	5	0	0
1	0	0	5	0	0	1	4.8
1	0	1	5	0	5	0	0
1	1	0	5	5	0	0	0
1	1	1	5	5	5	1	4.8

3.5 Basic gate level implementation of Adder

Half adder (as shown in Figure 3.1) is a combinational arithmetic circuit that adds two binary bits and produces a sum bit (S) and carry bit (C) as the output. If ‘A’ and ‘B’ are the input bits, then sum bit (S) is the XOR of inputs ‘A’ and ‘B’ and the carry bit (C) will be the AND of the inputs ‘A’ and ‘B’ (as shown in Figure 3.2). A half adder circuit can be easily constructed using one XOR gate and one AND gate. Half adder is the simplest of all adder circuit, but it has a major disadvantage. The half adder can add only two input bits (A and B) and has nothing to do with the carry if there is any in the input. So, if the input to a half adder has a carry, then it will be neglected it and adds only the A and B bits. That means the binary addition process is not complete and that’s why it is called a half adder. The truth table of a half adder is shown in Table 3.

Table 3.4 - Truth table for Half Adder

Input (A)	Input (B)	Carry (C)	SUM (S)
0	0	0	0
0	1	0	1
1	0	0	1

1	1	1	0

We can denote the first input bit as **A** and the second input bit as **B.** If **A** and **B** are added together, we should see the summation bit and carry bit. In the first three row **0 + 0, 0 + 1 or 1+ 0** the addition is **0 or 1** but there is no carry bit, but in the last row we added **1 + 1** and it is producing a carry bit of **1** along with result **0**.

So, if we see the operation of an adder circuit, we need only two inputs and it will produce two outputs, one is addition result, denoted as **SUM** and other one is **CARRY OUT** bit.

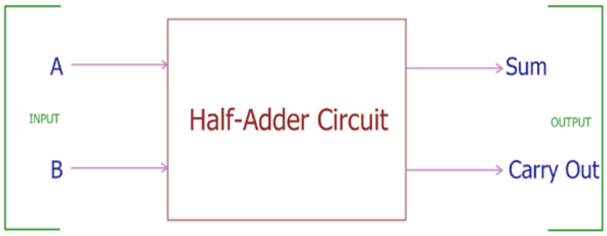


Figure 3.1 - Block diagram of a Half-Adder

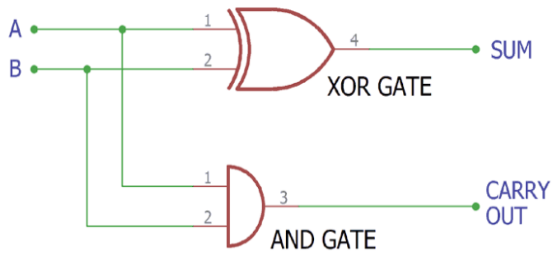


Figure 3.2 - Half Adder Circuit and its Construction

Task c

i. Procedure and observations

1. **Use the left side of the board to construct this circuit.** Circuit should be neat and compact, such that the left half side of the board is used.
2. Fix the required ICs which is under observation on the half separator line of breadboard, so there are no short connections.
3. Connect the wire to the main voltage source (Vcc = 5 V) whose another end is connected to last pin of the IC (14th pin from the notch).
4. Connect a ground of the IC (7^{th} pin from the notch) to the ground terminal provided by the DC power supply.
5. Make the connections according to Figure 3.3 and use **DIP Switch (8-pin)** to give inputs.
6. Connect output pins to LEDs.
7. Switch on the power supply.
8. The outputs of circuit shown in Figure 3.3 should satisfy the truth table shown in Table 3.
9. Complete
10.
11.
12. Table 3.5 with two inputs A and B as two bits.

ii. Make sure you do not remove the components from this board, save it for next time you will continue working on the same circuit, write your Names and IDs on a piece of paper and stick it on your board.

iii. Construct same another half adder circuit on the right side of the board.

Table 3.5 - Data Table for Half Adder 1

Input (A)	Input (B)	Carry	SUM
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 3.6 - Data Table for Half Adder 2

Input (A)	Input (B)	Carry	SUM
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

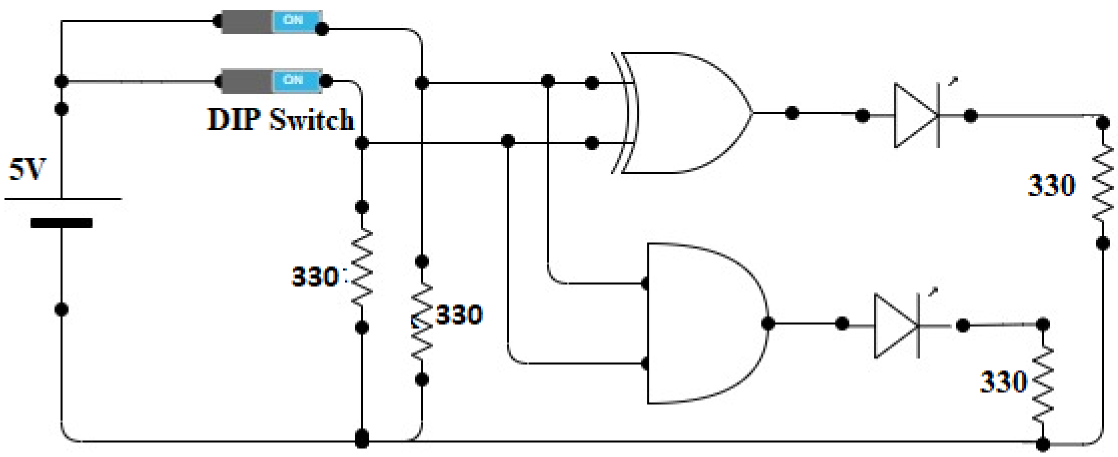


Figure 3.3 – Half Adder Circuit

3.6 Exercise

1. Draw complete circuit with missing Logic gates in Figure 3.4 (Hint: Refer to Boolean expressions at output to come up with gate)

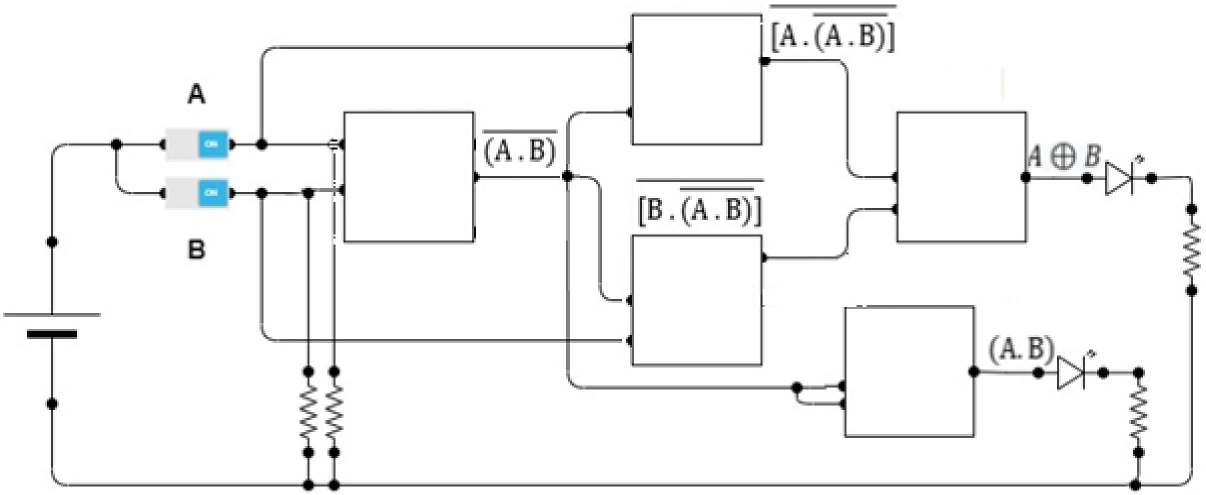


Figure 3.4 - Two input Logic circuit

2. Figure 3.5 shows XOR and its equivalent implementation using basic logic gates. Using the Table 3.6, verify that both circuits shown in figure 3.5, provides the same output

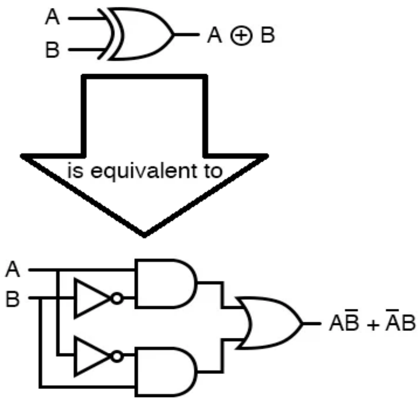


Figure 3.5 - XOR Equivalent Circuit

A	B	Output ($A \oplus B$)
0	0	0
0	1	1
1	0	1
1	1	0

Table 3.6 - Data Table

3. For the given Boolean Expressions:

$$S = C_i \oplus A \oplus B;$$
$$C_o = AB + (A \oplus B)\, C_i$$

a. Complete the circuit shown in Figure 3.6. (Hint: Use Boolean Expressions)
b. Complete the Data of Table 3.7 for Figure 3.6.

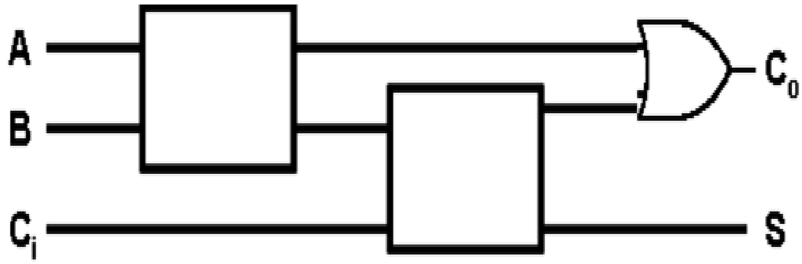
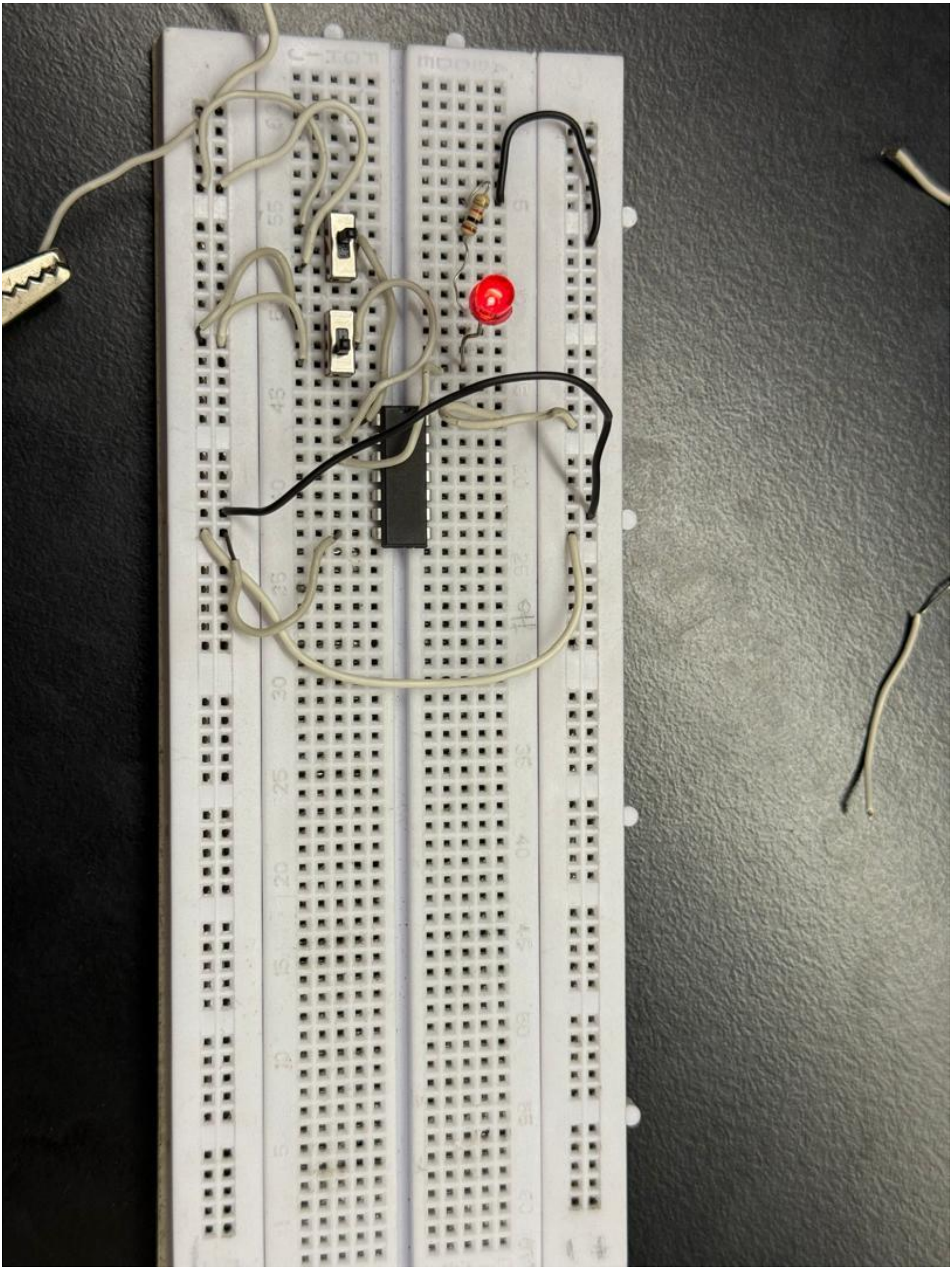


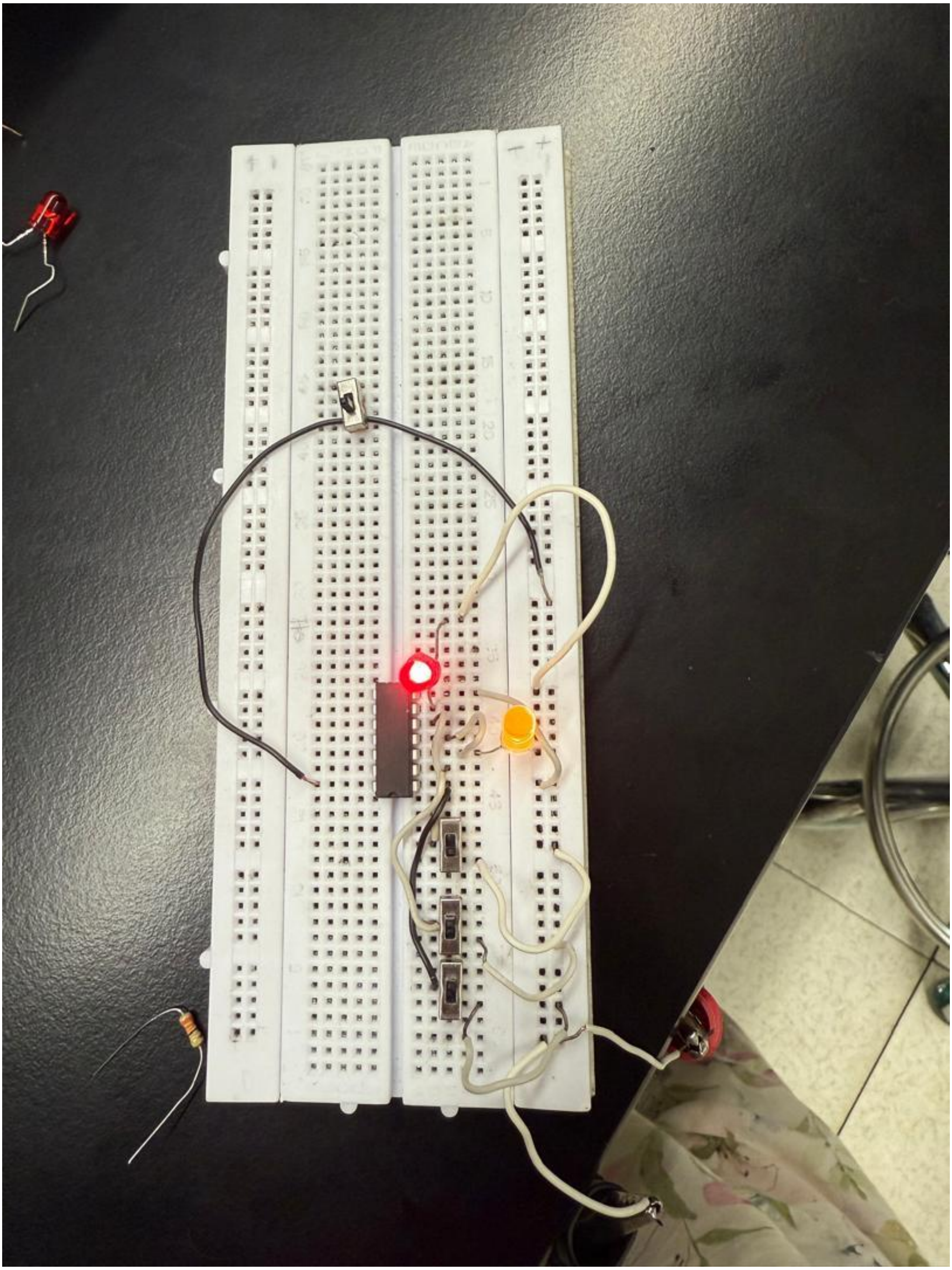
Figure 3.6 - Three input and two output Logic circuit.

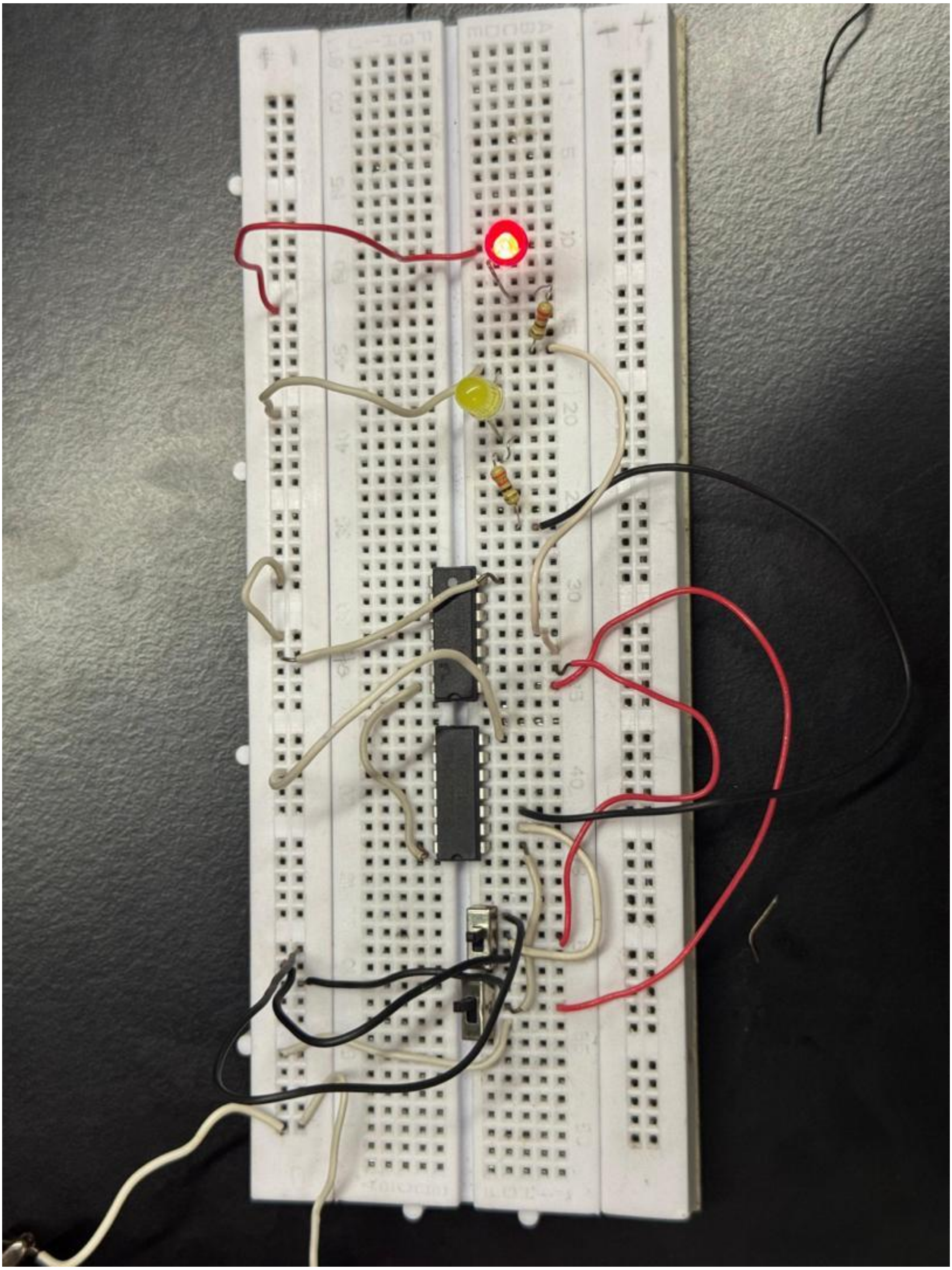
Table 3.7 - Data Table for circuit shown in Figure 3.6

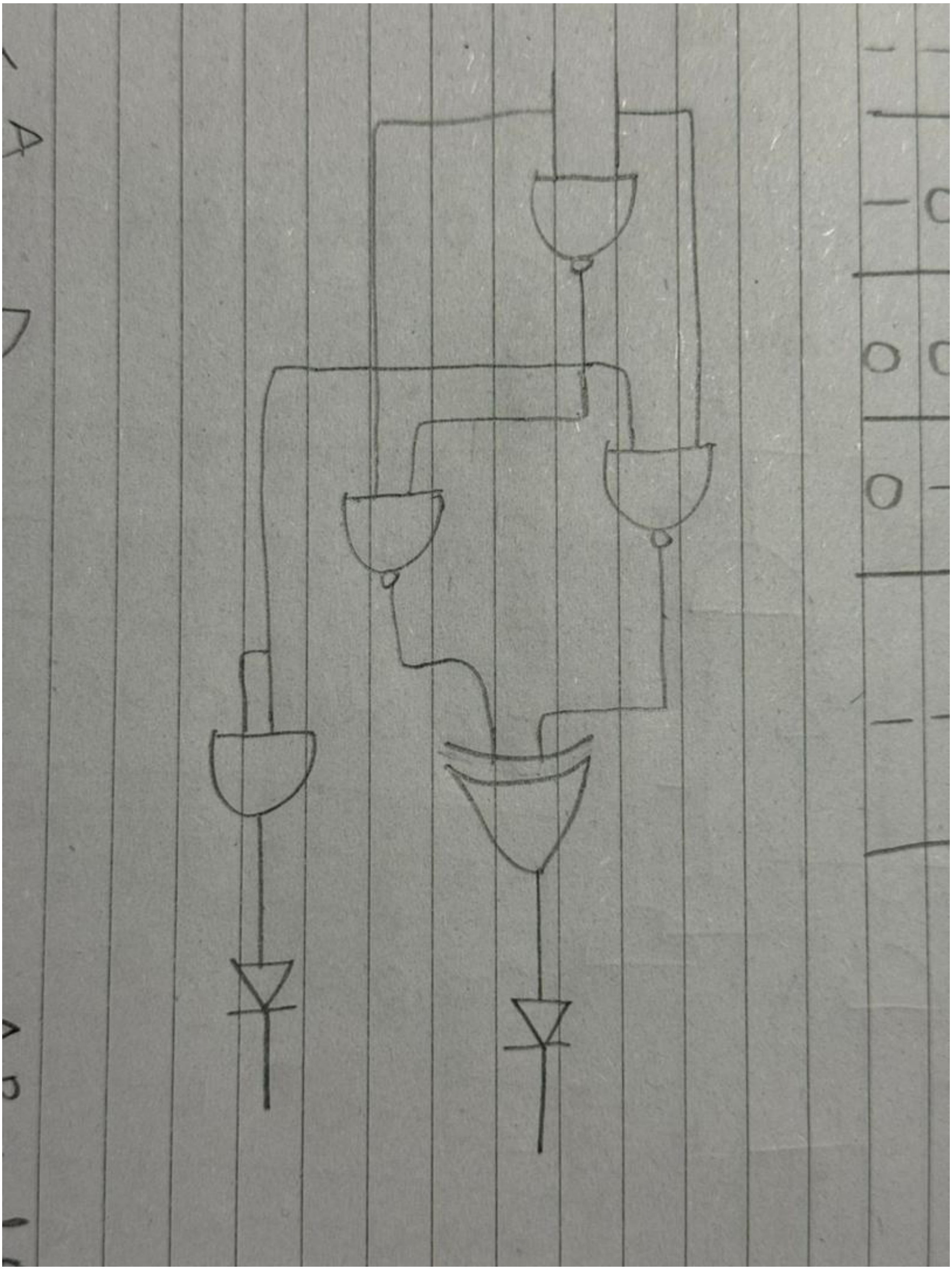
Ci	A	B	Co	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	0

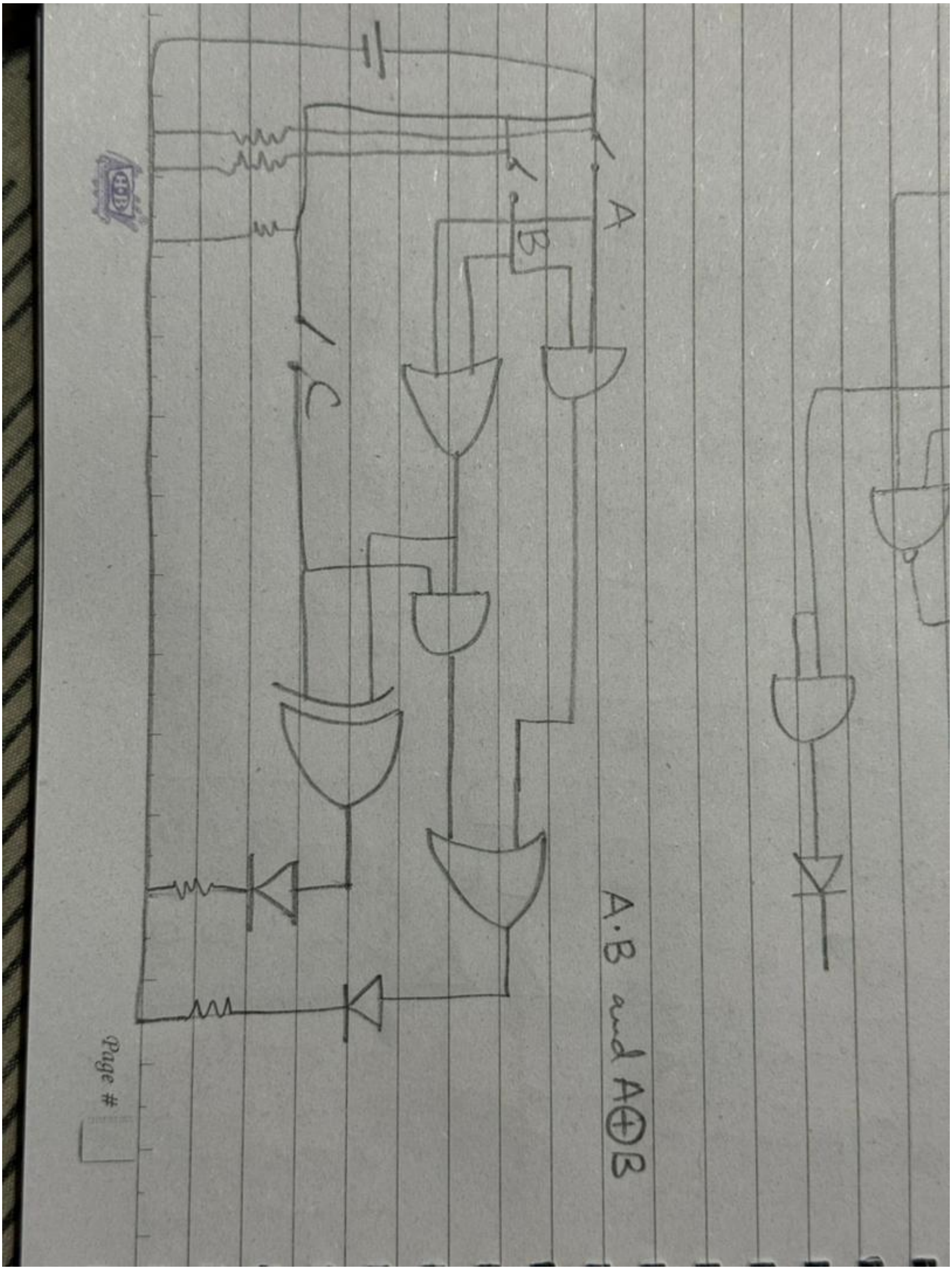
4. The logic circuit shown in exercise question 2 represents a full adder. Can you develop a logic where we can use more than one half adder to create a full adder? (If any extra sheet is required for working use A4 paper.)







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Assessment Rubric
Lab 03
Logic Gates & Adders

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Marks Distribution:

	Task No.	LR1 Circuit Layout	LR4 Data Collection	LR5 Results	LR 11 Design	LR9 Report
In - Lab	Task a	-	/05	-	-	-
	Task b	/10	-	/10	-	-
	Task c	/10	-	/10	-	-
Post Lab	1	-	-	/05	-	/05
	2	-	-	/05	-	/05
	3	-	-	/05	-	/05
	4	-	-	/05	/10	/05
Total: 100		/20	/05	/40	/10	/20
CLO Mapping		CLO1				CLO4
Learning Domain Level		Psy - 4				Aff - 3

Affective Domain Rubric		Points	CLO Mapped	Learning Domain Level
AR 7	Report Submission and Viva/Performance	/5	CLO 4	Aff - 3

CLO	Total Points	Points Obtained
1	75	
4	25	
Total	**100**	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	**Circuit Layout**	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort shown in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	**Viva**	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR10	**Analysis**	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	**Design**	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.